

Mini E-Commerce Backend

Cart Module Blueprint

1. Cart Module Overview

Responsibilities:

- Add Product to Cart
- Get User Cart
- Update Product Quantity
- Remove Product
- Clear Cart
- Calculate Cart Total

2. Cart APIs

Add Product

Item	Value
Method	POST
Endpoint	/cart
Authorization	User
Response	201 Created

Request

```
{
  "productId": "65df....",
  "quantity": 2
}
```

Response

```
{
  "message": "Product added successfully"
}
```

Get Cart

Item	Value
Method	GET
Endpoint	/cart
Authorization	User
Response	200 OK

Update Quantity

Item	Value
------	-------

Method	PATCH
Endpoint	/cart/:productId
Authorization	User
Response	200 OK

```
Request
{
  "quantity":5
}
```

Remove Product

Item	Value
Method	DELETE
Endpoint	/cart/:productId
Authorization	User
Response	200 OK

Clear Cart

Item	Value
Method	DELETE
Endpoint	/cart
Authorization	User
Response	200 OK

3. Cart Request Flow

Client → JWT Middleware → Cart Controller → Find User Cart → Modify Cart → MongoDB → Response

4. Cart Operations

Add Product

- Check Product Exists
- Check Quantity
- Add Item

Update Quantity

- Find Product
- Update Quantity

Remove Product

- Remove Selected Item

Clear Cart

- Delete All Cart Items

Get Cart

- Return All Cart Items

5. Business Rules

- User can only access his own cart.
- One cart for each user.
- If the product already exists: Increase quantity and do not insert duplicate product.
- Quantity must be greater than zero.
- Product must exist before adding.

6. Folder Structure

```
controllers/  
  cart.controller.js
```

```
routes/  
  cart.route.js
```

7. Controller Responsibilities

Function	Responsibility
addToCart	Add Product
getCart	Return User Cart
updateQuantity	Update Product Quantity
removeProduct	Delete Product
clearCart	Empty Cart

8. HTTP Status Codes

Code	Description
200	Success
201	Product Added
400	Validation Error
401	Unauthorized
404	Product Not Found
500	Internal Server Error

9. Common Error Scenarios

Error	Status Code
Product Not Found	404

Cart Not Found	404
Invalid Quantity	400
Unauthorized	401

10. Testing Checklist

- Add Product
- Update Quantity
- Remove Product
- Get Cart
- Clear Cart
- Duplicate Product
- Invalid Quantity
- JWT Authentication

11. Interview Notes

Why do we use one cart per user?

To keep all selected products in one place until checkout.

Why don't we create duplicate products in the cart?

Instead of creating another item, we increase the quantity.

Why do we validate quantity?

To prevent invalid values like zero or negative numbers.

Why does the user access only his own cart?

To protect users' data and maintain privacy.

What happens if the product doesn't exist?

404 Not Found

Why is the cart cleared after checkout?

Because the products have already been converted into an order.